Exercise 9 - solution

Paul Blanche

Exercise A

Question 1

We first load the data and look at the "summary", as always.

```
rm(list=ls())
load(url("http://paulblanche.com/files/colon2.rda"))
d <- colon2
summary(d)
```

```
##        id             rx            sex              age            obstruct
##  Min.   :  1   Obs    :315   Min.   :0.000   Min.   :18.00   Min.   :0.0000
##  1st Qu.:233   Lev    :310   1st Qu.:0.000   1st Qu.:53.00   1st Qu.:0.0000
##  Median :465   Lev+5FU:304   Median :1.000   Median :61.00   Median :0.0000
##  Mean   :465                 Mean   :0.521   Mean   :59.75   Mean   :0.1938
##  3rd Qu.:697                 3rd Qu.:1.000   3rd Qu.:69.00   3rd Qu.:0.0000
##  Max.   :929                 Max.   :1.000   Max.   :85.00   Max.   :1.0000
##
##      perfor            adhere           nodes           differ
##  Min.   :0.00000   Min.   :0.0000   Min.   : 0.00   Min.   :1.000
##  1st Qu.:0.00000   1st Qu.:0.0000   1st Qu.: 1.00   1st Qu.:2.000
##  Median :0.00000   Median :0.0000   Median : 2.00   Median :2.000
##  Mean   :0.02906   Mean   :0.1453   Mean   : 3.66   Mean   :2.063
##  3rd Qu.:0.00000   3rd Qu.:0.0000   3rd Qu.: 5.00   3rd Qu.:2.000
##  Max.   :1.00000   Max.   :1.0000   Max.   :33.00   Max.   :3.000
##                                     NA's   :18      NA's   :23
##      extent           surg            node4             time
##  Min.   :1.000   Min.   :0.0000   Min.   :0.0000   Min.   :   8
##  1st Qu.:3.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 370
##  Median :3.000   Median :0.0000   Median :0.0000   Median :1548
##  Mean   :2.887   Mean   :0.2659   Mean   :0.2745   Mean   :1405
##  3rd Qu.:3.000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:2289
##  Max.   :4.000   Max.   :1.0000   Max.   :1.0000   Max.   :3329
```

```
##
##      status           timeD         statusD           timeR
##  Min.   :0.0000   Min.   :  23   Min.   :0.0000   Min.   :   8
##  1st Qu.:0.0000   1st Qu.: 806   1st Qu.:0.0000   1st Qu.: 370
##  Median :1.0000   Median :1976   Median :0.0000   Median :1548
##  Mean   :0.5856   Mean   :1670   Mean   :0.4865   Mean   :1405
##  3rd Qu.:1.0000   3rd Qu.:2364   3rd Qu.:1.0000   3rd Qu.:2289
##  Max.   :2.0000   Max.   :3329   Max.   :1.0000   Max.   :3329
##
##     statusR
##  Min.   :0.0000
##  1st Qu.:0.0000
##  Median :1.0000
##  Mean   :0.5038
##  3rd Qu.:1.0000
##  Max.   :1.0000
##
```

Question 2

We keep only the observations corresponding to patients of these two groups.

```r
d <- d[which(d$rx %in% c("Obs","Lev+5FU")),]
d$rx <- droplevels(d$rx)   # drop the level no longer present in the dataset
```

We now look at the number of patients included in each treatment group.

```r
table(d$rx)
```

```
##
##     Obs Lev+5FU
##     315     304
```

We read that 315 did not receive the treatment and 304 did.

Question 3

Before creating the table, we first create factor variables for all the categorical variables and use appropriate labels.

```r
d$sex <- factor(d$sex,levels=c(1,0),labels=c("male","female"))
d$obstruct <- factor(d$obstruct,levels=c(0,1),labels=c("no","yes"))
d$perfor <- factor(d$perfor,levels=c(0,1),labels=c("no","yes"))
d$adhere <- factor(d$adhere,levels=c(0,1),labels=c("no","yes"))
d$differ <- factor(d$differ,levels=c(1,2,3),labels=c("well","moderate","poor"))
```

```r
d$extent <- factor(d$extent,levels=c(1,2,3,4),labels=c("submucosa",
                                                       "muscle",
                                                       "serosa",
                                                       "contiguous structures"))
d$surg <- factor(d$surg,levels=c(0,1),labels=c("short","long"))
d$node4 <- factor(d$node4,levels=c(0,1),labels=c("<=4",">4"))
```

We are now better prepared to create a descriptive table. The aim of this table is to summarize the baseline variable distributions in each treatment group. We report median, first and third quartile for quantitative variables and frequencies and percentages for categorical variables. As recommended by guidelines (e.g. CONSORT), we do not present useless p-values to test for differences between the two groups.

```r
library(Publish)
tab1 <- univariateTable(rx~Q(age) + Q(nodes) + sex +
                            obstruct + perfor + adhere +
                            differ + extent + surg + node4,
                        data=d,
                        compare.groups = FALSE,
                        show.totals = FALSE)
tab1
```

```
##    Variable                 Level Obs (n=315) Lev+5FU (n=304)
## 1       age          median [iqr] 60 [53, 68]     62 [52, 70]
## 2     nodes          median [iqr]    2 [1, 5]        2 [1, 4]
## 3                         missing           3               9
## 4       sex                  male  166 (52.7)      141 (46.4)
## 5                          female  149 (47.3)      163 (53.6)
## 6  obstruct                    no  252 (80.0)      250 (82.2)
## 7                             yes   63 (20.0)       54 (17.8)
## 8    perfor                    no  306 (97.1)      296 (97.4)
## 9                             yes     9 (2.9)         8 (2.6)
## 10   adhere                    no  268 (85.1)      265 (87.2)
## 11                            yes   47 (14.9)       39 (12.8)
## 12   differ                  well    27 (8.8)        29 (9.7)
## 13                        moderate  229 (74.4)      215 (72.1)
## 14                            poor   52 (16.9)       54 (18.1)
## 15                        missing           7               6
## 16   extent              submucosa     8 (2.5)        10 (3.3)
## 17                          muscle   38 (12.1)       32 (10.5)
## 18                          serosa  249 (79.0)      251 (82.6)
## 19           contiguous structures    20 (6.3)        11 (3.6)
## 20     surg                  short  224 (71.1)      228 (75.0)
## 21                           long   91 (28.9)       76 (25.0)
## 22    node4                    <=4  228 (72.4)      225 (74.0)
```

```
## 23                                >4   87 (27.6)       79 (26.0)
```

We see that the distributions of all the covariates seem to be fairly similar in the two groups. This is not at all a surprise: this was expected because of the randomization of the treatment allocation. Because of the random allocation to each treatment group, the patients are as old/young and healthy/commorbid in the two groups. The two populations differ only by treatment. Hence, if we observe a survival difference when comparing the two groups (at a later question), we can confidently conclude that this must be a consequence of the treatment.

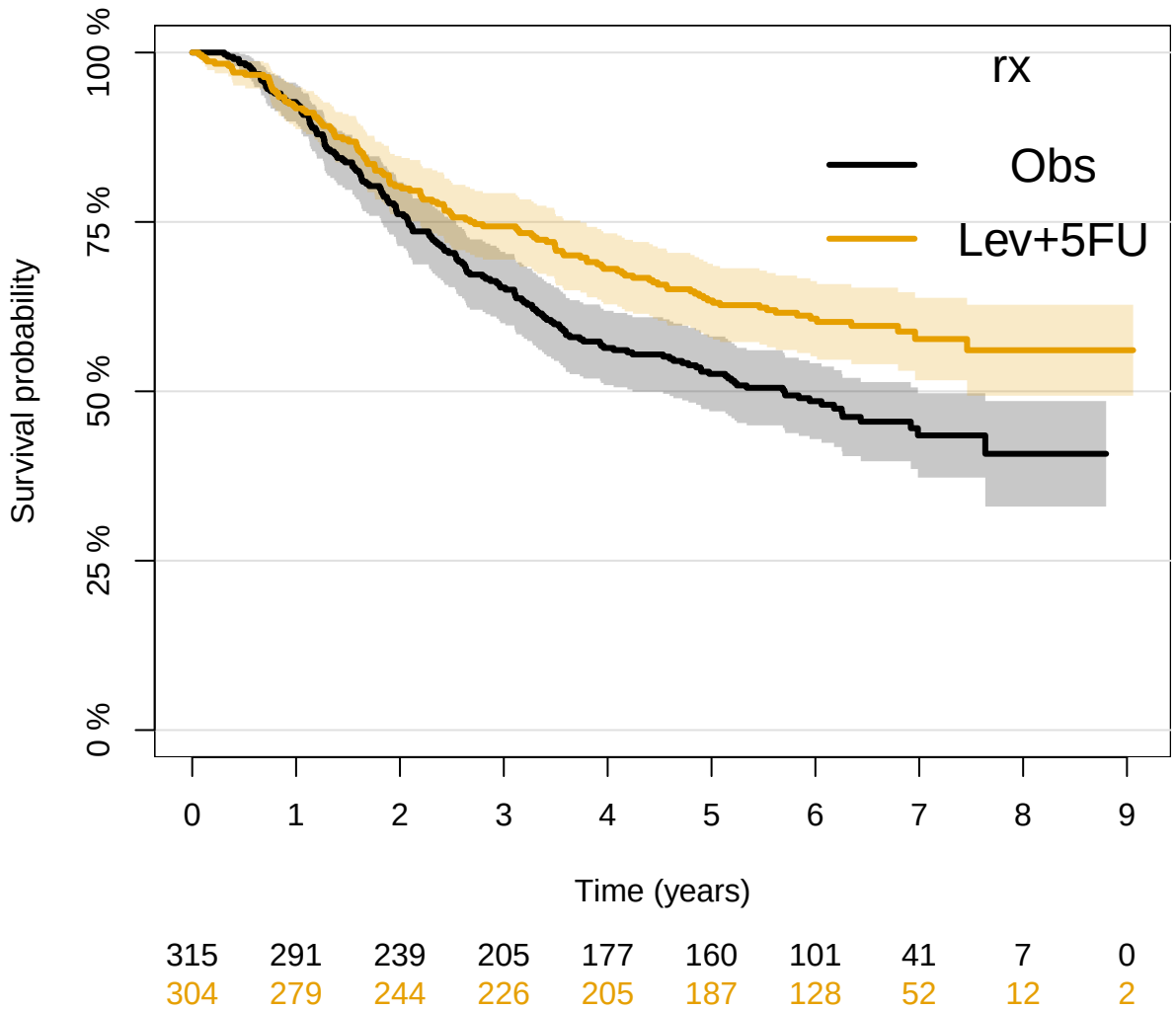
Question 4

We now use the Kaplan-Meier method to estimate the survival curves in each treatment group. We use the `timeD` and `statusD` variables, not `time` and `status`. This is because according to the description of the data, they are those relevant when studying all-cause death as the endpoint.

```r
library(survival)
library(prodlim)
KM1 <- prodlim(Surv(timeD,statusD)~rx,data=d)
```

We can then plot the curves. We use the option `timeconverter="days2years"` to show the years instead of days. The time unit is "days" in the dataset, but the interpretation is easier using "years". We could of course further fine tune the plot (if relevant) as exemplified in the R-demo.

```r
plot(KM1,timeconverter="days2years")
```



The plot suggests that the treatment improves the survival chances of the patients. Indeed, the survival curve corresponding to those treated is well above that of those not treated. This is the case everywhere (i.e. for all time points) maybe except at the very beginning (say within the first year), but there is anyway too much uncertainty there (as shown by the confidence intervals) to draw a clear conclusion. The (pointwise) 95% confidence intervals do not really cross after approximately 3 years. This suggests that the data contain rather strong evidence that the treatment work.

Question 5

We now compute the median survival times (with 95%-CI) for each treatment group.

```r
quantile(KM1)
```

```
## Quantiles of the event time distribution based on the Kaplan-Meier method.
##
## Table of quantiles and corresponding confidence limits:
##          rx     q quantile lower upper
```

```
##      <fctr> <num>    <num> <num> <num>
##  1:     Obs  0.00       NA    NA    NA
##  2:     Obs  0.25       NA    NA    NA
##  3:     Obs  0.50     2083  1548  2552
##  4:     Obs  0.75      760   665   928
##  5:     Obs  1.00      113   113   164
##  6: Lev+5FU  0.00       NA    NA    NA
##  7: Lev+5FU  0.25       NA    NA    NA
##  8: Lev+5FU  0.50       NA  2725    NA
##  9: Lev+5FU  0.75      993   802  1365
## 10: Lev+5FU  1.00       23    23    52
##
##
## Median with interquartile range (IQR):
##         rx        Median (IQR)
##     <fctr>              <char>
## 1:     Obs 2083.00 (760.00;NA)
## 2: Lev+5FU      NA (993.00;NA)
```

We estimate that, without treatment, half of the patients die within 2083 days (95%-CI=[1548, 2552]). We can convert to years as follows.

```r
round(c(2083,  1548,  2552)/365,1)
```

```
## [1] 5.7 4.2 7.0
```

That is 5.7 years (95%-CI=[4.2,7.0]). We could have read (approximately) this results from the Kaplan-Meier plot we previously produced. Indeed, we just need to read the x-values (i.e., the times) at which the survival curve and 95%-CI intersect the y=0.5 horizontal line (see lecture slides).

For the treated, we do not obtain an estimated value for the median. We obtain a value for the lower limit of the confidence interval only (2725 days, i.e. 7.5 years). Therefore, we can only conclude that we are 95% confident that the median survival time is larger than 7.5 years. This is because only the curve of the lower limit of the (pointwise) confidence interval intersect with the y=0.5 horizontal line. We do not observe enough deaths to know more about the median survival time for this group.

Question 6

We now report the estimated 7-year survival probabilities in each group, together with a 95%-CI.

```r
summary(KM1,time=7*365)
```

```
##        rx time n.risk n.event n.lost  surv se.surv lower upper
## 1     Obs 2555     41       0      0 0.435  0.0318 0.373 0.497
```

```
## 2 Lev+5FU 2555     52        0       0 0.577  0.0310 0.516 0.638
```

We read that we estimated the 7-year survival probability to be 43.5% (95%-CI=[37.3,49.7]) for those who did not receive the treatment, versus 57.7% (95%-CI=[51.6,63.8]) for those who did.

We now report the estimated difference in 7-year survival probability, together with a 95%-CI and a p-value.

```r
library(timeEL)
# 'group' variable, needs to be coded 0 or 1
d$group01 <- as.numeric(d$rx=="Obs")

ResDiffSurv <- TwoSampleKaplanMeier(time=d$timeD,
                                    status=d$statusD,
                                    group=d$group01,
                                    t=7*365,
                                    contr = list(method = "Wald"))

print(ResDiffSurv,what="Diff")
```

```
##
## Non-parametric inference to compare two risks (or survival probabilities) at time t=
##
## --------
## Wald-type inference for the Risk Difference:
##
## Estimate= 0.1422  and  95 % confidence interval= [ 0.0551 ; 0.2293 ]
##
## p-value= 0.0014  for 'H0: Diff= 0.0000 '.
```

We read that we estimate a statistically significant difference of 14.2% (95%-CI=[5.5,22.9], p=0.001) in favor of those receiving the treatment. By "statistically significant", we mean that the data contain enough evidence to reject the null hypothesis of no difference in 7-year survival probability.

Question 7

Before we move on to the main analysis (at question 11), we first compare the survival curves of the two treatment groups via a simple log-rank test (just to practice and explore the data further).

```r
ResLogRank <- survdiff(Surv(timeD,statusD)~rx,data=d)
print(ResLogRank,digits=6)
```

```
## Call:
## survdiff(formula = Surv(timeD, statusD) ~ rx, data = d)
```

```
##
##               N Observed Expected (O-E)^2/E (O-E)^2/V
## rx=Obs      315      168  141.117   5.12134   9.96567
## rx=Lev+5FU 304      123  149.883   4.82180   9.96567
##
##  Chisq= 10  on 1 degrees of freedom, p= 0.001595
```

We read that the p-value is significant (p=0.002). The null hypothesis of this test is that the survival curves of the two groups are identical (everywhere, i.e. at all times). The small p-value is not surprising, as we have previously seen that the curves are estimated to be very different, with (pointwise) 95%-CI which mostly do not overlap for time points between say 3 and 9 years (approximately).

Question 8

To accompany the p-value of the log-rank test, it is considered good practice to report an estimated "effect size" and 95%-CI. That is, the hazard ratio obtained from a univariate Cox model and its 95%-CI. We use the `coxph` function to compute this as follows.

```r
library(survival)
cox1 <- coxph(Surv(timeD,statusD)~rx,data=d)
summary(cox1)
```

```
## Call:
## coxph(formula = Surv(timeD, statusD) ~ rx, data = d)
##
##   n= 619, number of events= 291
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## rxLev+5FU -0.3728    0.6888   0.1188 -3.138   0.0017 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##           exp(coef) exp(-coef) lower .95 upper .95
## rxLev+5FU    0.6888      1.452    0.5457    0.8694
##
## Concordance= 0.542  (se = 0.015 )
## Likelihood ratio test= 9.98  on 1 df,   p=0.002
## Wald test            = 9.85  on 1 df,   p=0.002
## Score (logrank) test = 9.96  on 1 df,   p=0.002
```

We read that the hazard ratio is estimated to 0.69 (95%-CI=[0.55,0.87], p=0.002). Here the p-value corresponds to the null hypothesis of a hazard ratio equal to 1 (i.e. same hazard rate in both groups). We can therefore conclude that the hazard rate (i.e. the instantaneous risk of death among patients still alive at any time during the follow-up) is 0.69 times smaller

among the treated patients than among the patients who did not receive the treatment (95%-CI=[0.55,0.87], p=0.002).

We can also report the results in terms of a relative *instantaneous risk* (or rate) reduction, by computing "one minus the hazard ratio".

```
round((1-exp(cox1$coef))*100,1)
```

```
## rxLev+5FU
##      31.1
```

```
round(rev(1-exp(confint(cox1)))*100,1)
```

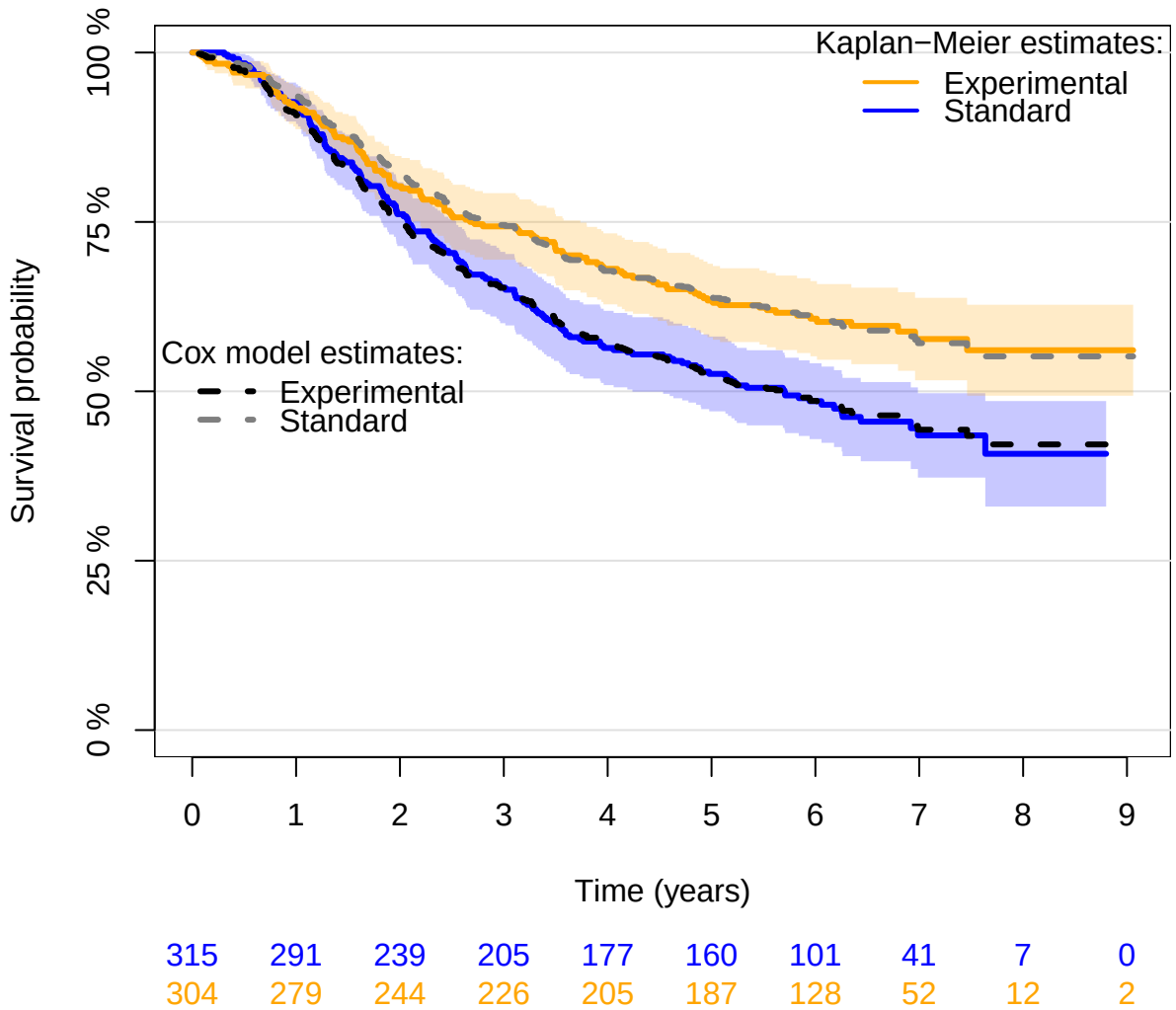
```
## [1] 13.1 45.4
```

Hence, we conclude that patients receiving fluorouracil plus levamisole had a 31.1% reduction in mortality rate (95%-CI=[13.1%,45.4%]). These results approximately match those of Moertel et al (1995), who analyzed an almost identical version of these data and found 33% (95%-CI=[16%,47%]).

Question 9

We now produce a simple plot to visually compare the survival curves for each treatment group, when they are estimated via either Kaplan-Meier or a univariate Cox model. This is to check the proportional hazards assumption of the Cox model.

```
# Step 1: create "new data" to predict the survival
dnew1 <- data.frame(rx="Obs")
dnew0 <- data.frame(rx="Lev+5FU")
# Step 2:  predict/estimate the survival probabilities for both groups
#  using the Cox model
scox1 <- survfit(cox1,newdata=dnew1)
scox0 <- survfit(cox1,newdata=dnew0)
# Step 3: plot Kaplan-Meier curves
plot(KM1,legend=FALSE,col=c("blue","orange"),timeconverter="days2years")
# Step 4: add the estimated survival curves from the Cox model
lines(scox0$time,scox0$surv,col="grey50",type="s",lwd=3,lty=2)
lines(scox1$time,scox1$surv,col="black",type="s",lwd=3,lty=2)
# Step 5: add legends
legend("left",legend=c("Experimental","Standard"),lwd=3,lty=2,
       col=c("black","grey50"),title="Cox model estimates:   ",bty="n")
legend("topright",legend=c("Experimental","Standard"),lwd=2,lty=1,
       col=c("orange","blue"),title="Kaplan-Meier estimates:",bty="n")
```



The curves estimated via both methods are very similar. We therefore conclude that nothing seems wrong with the proportional hazards assumption of the Cox model. Hence, we can further conclude that the hazard ratio that we have estimated via the Cox model is a sensible "summary" of the difference between the two survival curves. Had the model fit been bad, this would not have been a good "summary" of the difference.

Reminder: The Kaplan-Meier method does not make any assumption about how the curves differ from each other. By contrast, the Cox model does make such an assumption, via the proportional hazards assumption. For instance, the Cox model assumes that the two curves cannot cross (which is a consequence of the proportional hazards assumption).

Question 10

Before we move on to the main analysis (at question 11), we first compare the restricted mean survival times (RMST) at t = 7 years for the two treatment groups (just to practice and explore the data further).

We first need to create a 0/1 binary variable for the treatment group to use the `rmst2` function

of the package survRM2.

```
d$trt <- as.numeric(d$rx=="Lev+5FU")
table(d$trt)
```

```
##
##   0   1
## 315 304
```

We can now use the `rmst2` function as follows.

```
library(survRM2)
ResRMST <- rmst2(time=d$timeD/365, # trick to have a time unit in years
                 status=d$statusD,
                 arm=d$trt,
                 tau=7)            # Beware of the time unit!
ResRMST
```

```
##
## The truncation time: tau = 7  was specified.
##
## Restricted Mean Survival Time (RMST) by arm
##               Est.   se lower .95 upper .95
## RMST (arm=1) 5.190 0.14     4.917     5.464
## RMST (arm=0) 4.635 0.14     4.360     4.910
##
##
## Restricted Mean Time Lost (RMTL) by arm
##               Est.   se lower .95 upper .95
## RMTL (arm=1) 1.810 0.14     1.536     2.083
## RMTL (arm=0) 2.365 0.14     2.090     2.640
##
##
## Between-group contrast
##                       Est. lower .95 upper .95     p
## RMST (arm=1)-(arm=0) 0.556     0.168     0.944 0.005
## RMST (arm=1)/(arm=0) 1.120     1.034     1.212 0.005
## RMTL (arm=1)/(arm=0) 0.765     0.632     0.926 0.006
```

We estimate that, if we treat future patients from the same population with fluorouracil plus levamisole and follow them for 7 years, the average time spent alive will be approximately 5.2 years (95%-CI=[4.9,5.5]). By contrast, if we do not treat future patients from the same population and follow them for 7 years, the average time spent alive will be approximately 4.6 years (95%-CI=[4.4,4.9]). That is, 0.56 years less (95%-CI=[0.17,0.94], p=0.005).

Question 11

We now proceed to the main analysis. We fit a multiple Cox regression model to compare the survival chances of a patient who received levamisole plus fluorouracil to that of a patient who did not, when both patients are similar with regards to:

- obstruction of the colon by the tumor (`obstruct`),
- number of positive lymph nodes (more or less than 4, `node4`)
- time from surgery to inclusion into the trial (short vs long, `surg`)
- age and sex

We assume that the following was prespecified. We do not model any interaction and age is used in the model as a continuous variable with a (usual/simple) log-linear effect on the hazard rate.

```r
cox2 <- coxph(Surv(timeD,statusD)~rx+obstruct+node4+surg+age+sex,data=d)
summary(cox2)
```

```
## Call:
## coxph(formula = Surv(timeD, statusD) ~ rx + obstruct + node4 +
##     surg + age + sex, data = d)
##
##   n= 619, number of events= 291
##
##                  coef exp(coef)  se(coef)      z Pr(>|z|)
## rxLev+5FU   -0.360768  0.697141  0.119597 -3.017  0.00256 **
## obstructyes  0.152375  1.164597  0.146792  1.038  0.29926
## node4>4      0.937406  2.553348  0.121789  7.697 1.39e-14 ***
## surglong     0.241523  1.273187  0.127573  1.893  0.05833 .
## age          0.003651  1.003658  0.004848  0.753  0.45140
## sexfemale    0.090796  1.095046  0.118013  0.769  0.44167
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##             exp(coef) exp(-coef) lower .95 upper .95
## rxLev+5FU      0.6971     1.4344    0.5515    0.8813
## obstructyes    1.1646     0.8587    0.8734    1.5528
## node4>4        2.5533     0.3916    2.0111    3.2417
## surglong       1.2732     0.7854    0.9915    1.6349
## age            1.0037     0.9964    0.9942    1.0132
## sexfemale      1.0950     0.9132    0.8689    1.3800
##
## Concordance= 0.646  (se = 0.016 )
## Likelihood ratio test= 69.76  on 6 df,   p=5e-13
## Wald test            = 75.73  on 6 df,   p=3e-14
```

```
## Score (logrank) test = 80.09  on 6 df,   p=3e-15
```

We estimate the hazard ratio to be 0.70 (95%-CI=[0.55,0.88], p=0.003). That is, we estimate that the mortality rate is reduced by 30% (95%-CI=[12%,45%], p=0.003), for a treated patient, as compared to a patient who is not treated, when both have the same age and sex, and are also similar with respect to time from surgery to inclusion into the trial, number of positive lymph nodes and obstruction of the colon by the tumor. The main analysis shows significant results: we obtained a positive finding!

Question 12

One could wonder whether modeling a log-linear effect of age on the hazard rate was a good idea. It is, after all and to some extent, an arbitrary choice. It can be interesting to compare the previous conclusions to those obtained when modeling the effect of age via age groups, which does not require to assume a log-linear effect. For instance, we can use four age groups: 18-50, 50-60, 70-85.

```r
d$agec <- cut(d$age,breaks=c(18,50,60,70,85),include.lowest=TRUE)
table(d$agec,useNA="always")
```

```
##
## [18,50] (50,60] (60,70] (70,85]    <NA>
##     132     167     202     118       0
```

```r
cox2c <- coxph(Surv(timeD,statusD)~rx+obstruct+node4+surg+agec+sex,data=d)
summary(cox2c)
```

```
## Call:
## coxph(formula = Surv(timeD, statusD) ~ rx + obstruct + node4 +
##     surg + agec + sex, data = d)
##
##   n= 619, number of events= 291
##
##                 coef exp(coef) se(coef)      z Pr(>|z|)
## rxLev+5FU   -0.38714   0.67899  0.12043 -3.215  0.00131 **
## obstructyes  0.12979   1.13859  0.14685  0.884  0.37678
## node4>4      0.93300   2.54212  0.12179  7.661 1.85e-14 ***
## surglong     0.23194   1.26104  0.12759  1.818  0.06909 .
## agec(50,60] -0.21717   0.80479  0.17348 -1.252  0.21062
## agec(60,70]  0.14804   1.15956  0.15892  0.932  0.35157
## agec(70,85] -0.03359   0.96697  0.18601 -0.181  0.85672
## sexfemale    0.10476   1.11045  0.11871  0.883  0.37750
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##             exp(coef) exp(-coef) lower .95 upper .95
```

```
## rxLev+5FU      0.6790     1.4728    0.5362    0.8598
## obstructyes    1.1386     0.8783    0.8538    1.5183
## node4>4        2.5421     0.3934    2.0023    3.2275
## surglong       1.2610     0.7930    0.9820    1.6193
## agec(50,60]    0.8048     1.2426    0.5728    1.1307
## agec(60,70]    1.1596     0.8624    0.8492    1.5833
## agec(70,85]    0.9670     1.0342    0.6716    1.3924
## sexfemale      1.1104     0.9005    0.8799    1.4014
##
## Concordance= 0.65  (se = 0.016 )
## Likelihood ratio test= 74.72  on 8 df,   p=6e-13
## Wald test            = 80.38  on 8 df,   p=4e-14
## Score (logrank) test = 84.74  on 8 df,   p=5e-15
```

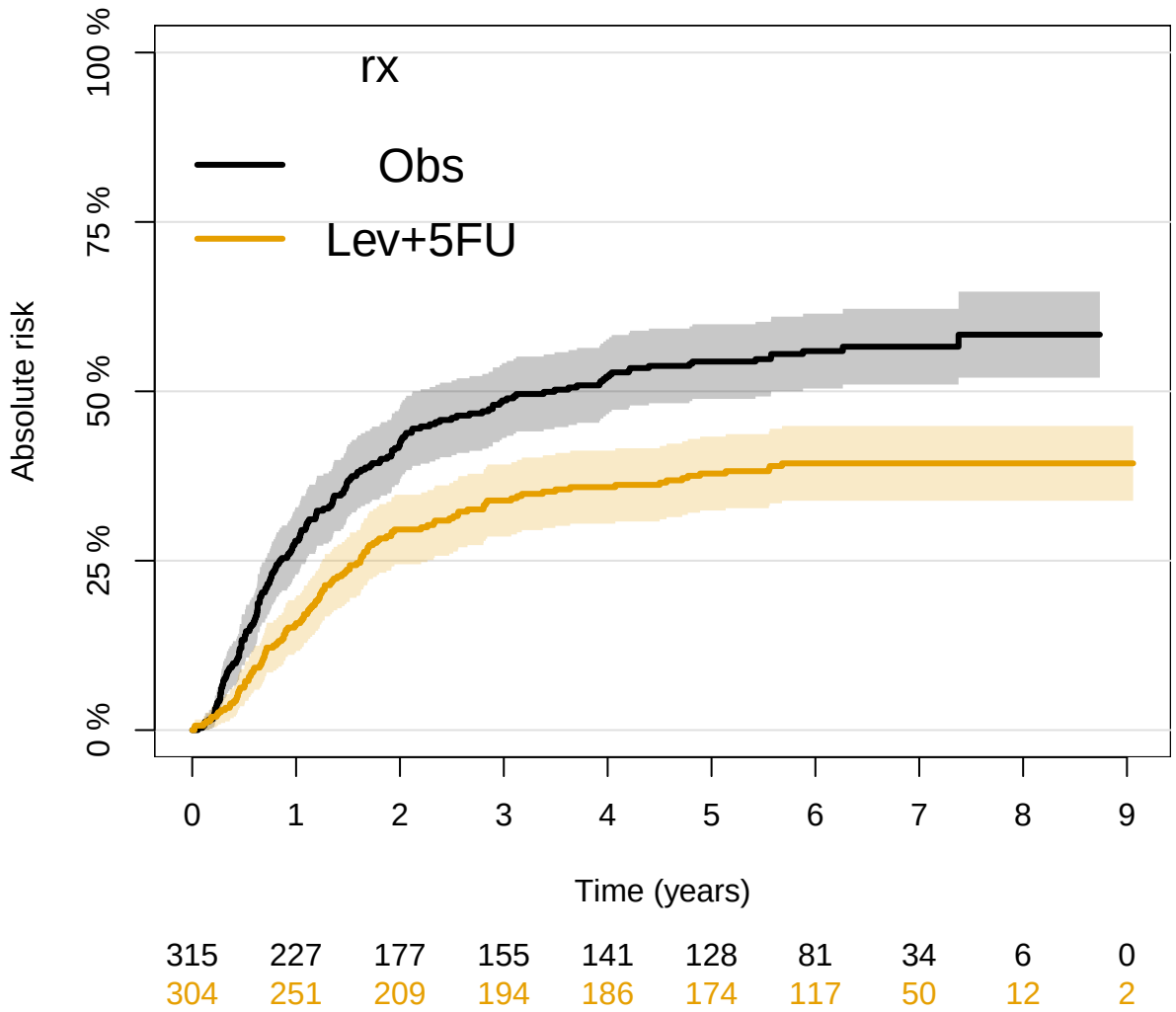
We now estimate the hazard ratio to be 0.68 (95%-CI=[0.54,0.86], p=0.001). The main conclusion does not change at all. This simple "sensitivity analysis" is very reassuring.

Exercise B

(only the R code is provided)

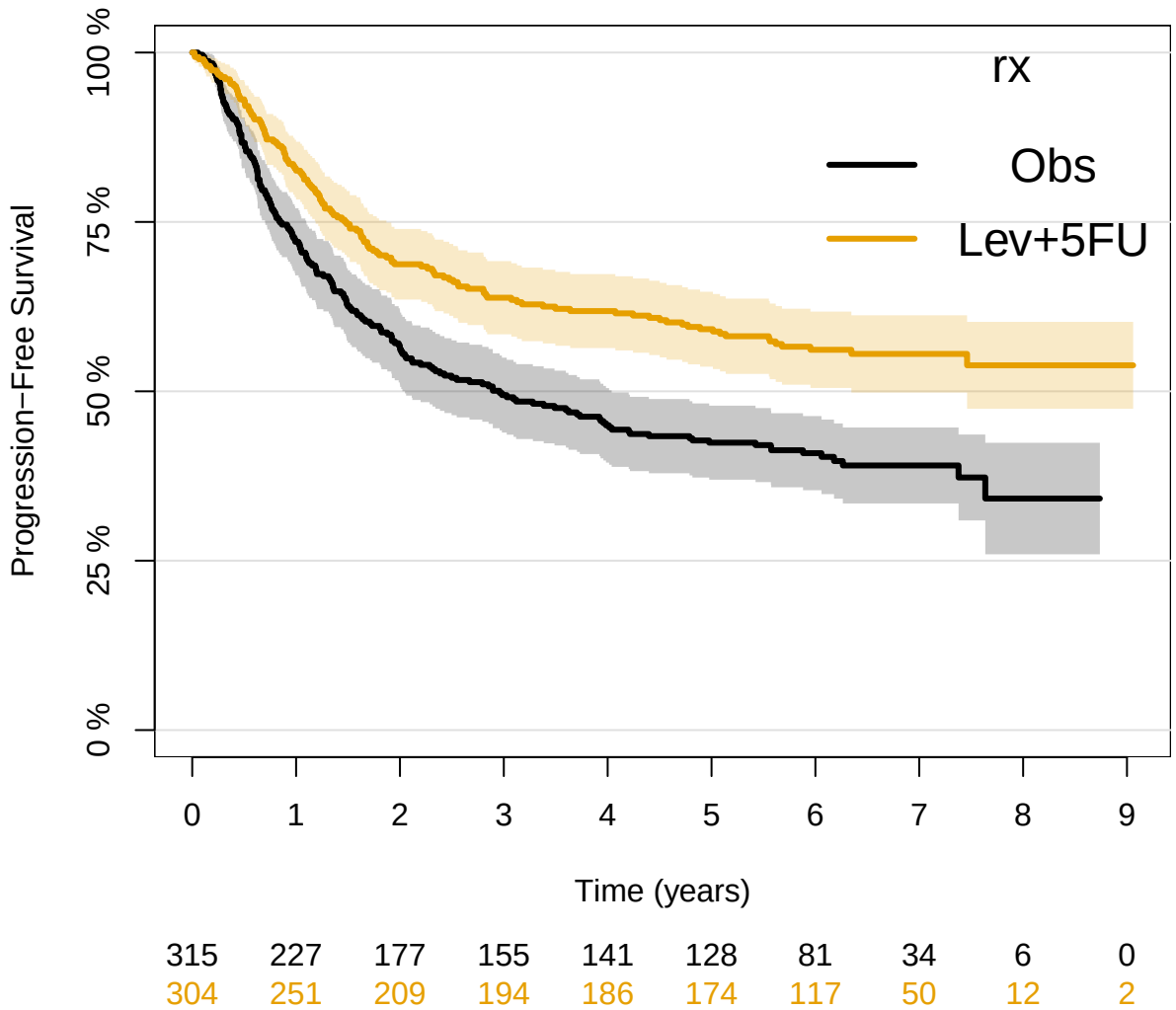
Question 1

```r
AJ1 <- prodlim(Hist(time,status)~rx,data=d)
plot(AJ1,timeconverter="days2years")
```



Question 2

```
KMPFS <- prodlim(Surv(time,status!=0)~rx,data=d)
plot(KMPFS,timeconverter="days2years",ylab="Progression-Free Survival")
```




```r
cox2PFS <- coxph(Surv(time,status!=0)~rx+obstruct+node4+surg+age+sex,data=d)
summary(cox2PFS)
```

```
## Call:
## coxph(formula = Surv(time, status != 0) ~ rx + obstruct + node4 +
##     surg + age + sex, data = d)
##
##   n= 619, number of events= 324
##
##                    coef  exp(coef)   se(coef)      z Pr(>|z|)
## rxLev+5FU   -0.4605591  0.6309308  0.1136788 -4.051 5.09e-05 ***
## obstructyes  0.1249044  1.1330401  0.1401069  0.891   0.3727
## node4>4      0.8466340  2.3317849  0.1162072  7.286 3.20e-13 ***
## surglong     0.2492774  1.2830979  0.1206781  2.066   0.0389 *
## age         -0.0003737  0.9996264  0.0046560 -0.080   0.9360
## sexfemale    0.1134394  1.1201240  0.1118273  1.014   0.3104
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##              exp(coef) exp(-coef) lower .95 upper .95
## rxLev+5FU      0.6309     1.5850    0.5049    0.7884
## obstructyes    1.1330     0.8826    0.8610    1.4911
## node4>4        2.3318     0.4289    1.8568    2.9282
## surglong       1.2831     0.7794    1.0128    1.6255
## age            0.9996     1.0004    0.9905    1.0088
## sexfemale      1.1201     0.8928    0.8997    1.3946
##
## Concordance= 0.647  (se = 0.015 )
## Likelihood ratio test= 73.57  on 6 df,   p=8e-14
## Wald test            = 78.85  on 6 df,   p=6e-15
## Score (logrank) test = 82.5  on 6 df,   p=1e-15
```